Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the trigonometric equation $\cos(2\theta) + 3\sin\theta = 2$ for all solutions $\theta \in [0, 2\pi)$.

Substitute $\cos(2\theta) = 1 - 2\sin^2\theta$ to express the entire equation in terms of $\sin\theta$: $(1 - 2\sin^2\theta) + 3\sin\theta = 2$. $-2\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 2\sin^2\theta - 3\sin\theta + 1 = 0$. Factor the quadratic in $\sin\theta$: $(2\sin\theta - 1)(\sin\theta - 1) = 0$. Case 1: $2\sin\theta - 1 = 0 \Rightarrow \sin\theta = \frac{1}{2}$. On $[0, 2\pi)$, $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$. Case 2: $\sin\theta - 1 = 0 \Rightarrow \sin\theta = 1$. On.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Verify the trigonometric identity: $\frac{\sin(2x)}{\sin x} - \frac{\cos(2x)}{\cos x} = \sec x$.

Expand the double-angle expressions on the left-hand side: LHS $= \frac{2\sin x\cos x}{\sin x} - \frac{2\cos^2 x - 1}{\cos x}$. Cancel $\sin x$ in the first term and combine over a common denominator of $\cos x$: $= 2\cos x - \frac{2\cos^2 x - 1}{\cos x} = \frac{2\cos^2 x - (2\cos^2 x - 1)}{\cos x}$. $= \frac{2\cos^2 x - 2\cos^2 x + 1}{\cos x} = \frac{1}{\cos x} = \sec x =$ RHS. ✓

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a circle of radius 10 cm, chord AB subtends an angle of $\theta = \frac{2\pi}{3}$ radians at the centre O. Find: (a) the exact length of chord AB, (b) the exact area of the minor segment bounded by chord AB and minor arc AB, in terms of π and $\sqrt{3}$.

(a) Use the cosine rule in triangle OAB with $OA = OB = 10$ and $\angle AOB = \frac{2\pi}{3} = 120^\circ$: $AB^2 = 10^2 + 10^2 - 2(10)(10)\cos\left(\frac{2\pi}{3}\right) = 100 + 100 - 200\left(-\frac{1}{2}\right) = 200 + 100 = 300$ Thus $AB = \sqrt{300} = 10\sqrt{3}$ cm. (b) Area of sector OAB: $A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{1}{2}(10^2)\left(\frac{2\pi}{3}\right) = \frac{100\pi}{3}$ cm^2 Area of triangle OAB.

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5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A small particle of weight 24 N is held in equilibrium by two light inextensible strings. String A makes an angle of 30° with the upward vertical, and string B makes an angle of 60° with the upward vertical. Find the tensions T_A and T_B in the two strings.

Notice that the angle between string A and string B is $30^\circ + 60^\circ = 90^\circ$, so the strings are perpendicular. Resolve horizontally: $T_A \sin 30^\circ = T_B \sin 60^\circ \Rightarrow T_A(0.5) = T_B\left(\frac{\sqrt{3}}{2}\right) \Rightarrow T_A = \sqrt{3}T_B$ Resolve vertically: $T_A \cos 30^\circ + T_B \cos 60^\circ = 24 \Rightarrow T_A\left(\frac{\sqrt{3}}{2}\right) + T_B(0.5) = 24$ Substitute $T_A = \sqrt{3}T_B$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 2 · MediumEvaluate the trigonometric limit: $\lim_{x \to 0} \frac{1 - \cos(3x)}{x^2}$.

Multiply numerator and denominator by the conjugate $1 + \cos(3x)$: $\lim_{x \to 0} \frac{(1 - \cos(3x))(1 + \cos(3x))}{x^2(1 + \cos(3x))} = \lim_{x \to 0} \frac{1 - \cos^2(3x)}{x^2(1 + \cos(3x))} = \lim_{x \to 0} \frac{\sin^2(3x)}{x^2(1 + \cos(3x))}$. Re-group terms to leverage $\lim_{u \to 0} \frac{\sin u}{u} = 1$: $= \lim_{x \to 0} \left[\left(\frac{\sin(3x)}{3x}\right)^2 \cdot 3^2 \cdot \frac{1}{1 + \cos(3x)}\right]$. Evaluating individual limits.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

Step 1: Factor the quadratic in $\sin x$: $(2\sin x - 1)(\sin x - 1) = 0$. Step 2: Solve each factor branch:Branch 1: $2\sin x - 1 = 0 \Rightarrow \sin x = \frac{1}{2}$. On $[0, 2\pi)$, $x = \frac{\pi}{6}$ (Quadrant I) or $x = \frac{5\pi}{6}$ (Quadrant II).Branch 2: $\sin x - 1 = 0 \Rightarrow \sin x = 1$. On $[0, 2\pi)$, $x = \frac{\pi}{2}$.The complete solution set is $x \in \left\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}\right\}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, $a = 8$, $b = 11$, and $A = 35^\circ$. Determine the number of possible triangles and find the measure of angle B for all valid cases to the nearest tenth of a degree.

Calculate the altitude $h = b \sin A = 11 \sin 35^\circ \approx 11(0.5736) = 6.31$. Since $h two possible triangles. By the Law of Sines$: \sin B = \frac{b\sin A}{a} = \frac{6.31}{8} \approx 0.7887.x. Second solution (obtuse): $B_2 = 180^\circ - 52.1^\circ = 127.9^\circ$. Check validity: $A + B_2 = 35^\circ + 127.9^\circ = 162.9^\circ 0$. Both $B_1 \approx 52.1^\circ$ and $B_2 \approx 127.9^\circ$ are valid.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A circle has radius 8 cm. A sector of this circle has a total perimeter of 28 cm. Find the angle of the sector in radians, and calculate the exact area of the sector.

The perimeter of a sector is given by $P = s + 2r = r\theta + 2r$. Substitute $P = 28$ and $r = 8$: $28 = 8\theta + 2(8) \Rightarrow 28 = 8\theta + 16 \Rightarrow 8\theta = 12 \Rightarrow \theta = \frac{12}{8} = 1.5$ radians The area of the sector is: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(8^2)(1.5) = \frac{1}{2}(64)(1.5) = 32 \times 1.5 = 48$ cm^2

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A surveyor measures distances from an observation point C to two landmarks A and B: $AC = 140$ m and $BC = 210$ m, with included angle $\angle ACB = 78^\circ$. Find the distance AB between the landmarks to the nearest tenth of a meter, and compute the area of ΔABC to the nearest square meter.

Apply the Law of Cosines to find side $c = AB$: $c^2 = a^2 + b^2 - 2ab\cos C = 210^2 + 140^2 - 2(210)(140)\cos 78^\circ$. $c^2 = 44100 + 19600 - 58800(0.2079) = 63700 - 12225 = 51475$. $c = \sqrt{51475} \approx 226.9$ m. Compute the area: Area $= \frac{1}{2}ab\sin C = \frac{1}{2}(210)(140)\sin 78^\circ = 14700(0.9781) \approx 14379$ m^2.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

[Functions: Composite & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find the center, radius, and standard Cartesian equation of the circle given by the general equation $x^2 + y^2 - 8x + 6y - 11 = 0$.

Step 1: Group variables and complete the square: $(x^2 - 8x) + (y^2 + 6y) = 11$. $(x^2 - 8x + 16) + (y^2 + 6y + 9) = 11 + 16 + 9 = 36$. $(x - 4)^2 + (y + 3)^2 = 36$. Step 2: Extract circle invariants: Center $(h, k) = (4, -3)$ and radius $r = \sqrt{36} = 6$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · AdvancedEvaluate the limit analytically without using L'Hôpital's Rule: $\lim_{x \to 0} \frac{\sqrt{1 + \tan x} - \sqrt{1 + \sin x}}{x^3}$.

Step 1 · Rationalize the Numerator: Multiply by $\sqrt{1 + \tan x} + \sqrt{1 + \sin x}$: $\lim_{x \to 0} \frac{(1 + \tan x) - (1 + \sin x)}{x^3(\sqrt{1 + \tan x} + \sqrt{1 + \sin x})} = \lim_{x \to 0} \frac{\tan x - \sin x}{x^3(\sqrt{1 + \tan x} + \sqrt{1 + \sin x})}$. Step 2 · Factor Trigonometric Terms: $\tan x - \sin x = \frac{\sin x}{\cos x} - \sin x = \sin x\left(\frac{1 - \cos x}{\cos x}\right)$. Multiply numerator and denominator by $1 + \cos x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Level 3 · AdvancedFind the coordinates of all points on the curve $x^2 - xy + y^2 = 9$ where the tangent line is horizontal.

Step 1 · Implicit Differentiation: Differentiate with respect to x: $2x - \left(1 \cdot y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0 \Rightarrow 2x - y + (2y - x)\frac{dy}{dx} = 0$. Solve for $\frac{dy}{dx}$: $\frac{dy}{dx} = \frac{y - 2x}{2y - x}$. Step 2 · Condition for Horizontal Tangent: Set the numerator equal to zero while ensuring the denominator is non-zero: $y - 2x = 0 \Rightarrow y = 2x$. Step 3 · Substitute into Original Curve.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

A block of mass $5\ \text{kg}$ is released from rest at the top of a rough plane inclined at 30° to the horizontal. The length of the incline is $8\ \text{m}$ and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8\ \text{m/s}^2$, calculate the speed of the block at the bottom of the incline.

1. Vertical height descended: $h = 8\sin 30^\circ = 8(0.5) = 4\ \text{m}$. Initial gravitational potential energy: $\text{PE} = mgh = 5(9.8)(4) = 196\ \text{J}$ 2. Normal reaction: $R = mg\cos 30^\circ = 5(9.8)\left(\frac{\sqrt{3}}{2}\right) = 24.5\sqrt{3}\ \text{N}$. Friction force: $F = \mu R = 0.2(24.5\sqrt{3}) = 4.9\sqrt{3}\ \text{N}$. Work done against friction over distance $d = 8\ \text{m}$: $W_{\text{fric}} = F \times d = (4.9\sqrt{3}) \times 8 = 39.2\sqrt{3} \approx 67.896\ \text{J}$ 3. By.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Level 3 · AdvancedLet $h(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Prove that h is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that h is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

Step 1 · Algebraic Monotonicity: Rewrite $h(x)$ by dividing numerator and denominator by e^x: $h(x) = \frac{1 - e^{-2x}}{1 + e^{-2x}} = \frac{(1 + e^{-2x}) - 2e^{-2x}}{1 + e^{-2x}} = 1 - \frac{2}{e^{2x} + 1}$. As x, $\frac{2}{e^{2x} + 1} \in (0, 2) \Rightarrow h(x) = 1 - \frac{2}{e^{2x} + 1} \in (-1, 1)$. The domain of h^{-1} is $(-1, 1)$. Set $y = \frac{e^{2x} - 1}{e^{2x} + 1}$: $y(e^{2x} + 1) = e^{2x} - 1 \Rightarrow ye^{2x} + y = e^{2x} - 1 \Rightarrow e^{2x}(1 - y) = 1 + y$. For $y \in (-1, 1)$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

Step 1: Compute dot product and squared magnitude: $\mathbf{u} \cdot \mathbf{v} = (6)(3) + (2)(4) = 18 + 8 = 26$. $||\mathbf{v}||^2 = 3^2 + 4^2 = 9 + 16 = 25$. Step 2: Vector projection (parallel component): $\mathbf{u}_{\parallel} = \text{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v} = \frac{26}{25}\langle 3, 4 \rangle = \left\langle \frac{78}{25}, \frac{104}{25} \right\rangle = \langle 3.12, 4.16 \rangle$. Step 3: Orthogonal component.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.02 The Poisson Distribution

Level 2 · MediumFind the point (x, y) on the parabola $y = x^2$ that is closest to the point $(0, 2)$.

Step 1 · Distance Objective: The distance d from (x, x^2) to $(0, 2)$ is: $d = \sqrt{(x-0)^2 + (x^2-2)^2}$. Minimizing d is equivalent to minimizing the squared distance function $S(x) = d^2$:

$S(x) = x^2 + (x^2-2)^2 = x^2 + x^4 - 4x^2 + 4 = x^4 - 3x^2 + 4$. Step 2 · Critical Numbers:

$S'(x) = 4x^3 - 6x = 2x(2x^2 - 3) = 0$. Critical roots: $x = 0$ and $x = \pm\sqrt{\frac{3}{2}} = \pm\frac{\sqrt{6}}{2}$. Step 3 · Second Derivative Test: $S''(x) = 12x^2 - 6$. • At $x = 0$: $S''(0) = -60$ (certified loc.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Further Integration Techniques

A stone is thrown vertically upwards with speed 15 m/s from the top of a cliff 40 m above sea level. Taking $g = 9.8$ m/s^2 and upwards as positive: (a) find the maximum height reached above the cliff, (b) calculate the time taken for the stone to reach the sea.

(a) At maximum height, $v = 0$. $u = +15$ m/s, $a = -9.8$ m/s^2:

$v^2 = u^2 + 2as \Rightarrow 0 = 15^2 + 2(-9.8)s \Rightarrow 19.6s = 225 \Rightarrow s = \frac{225}{19.6} \approx 11.48 \text{ m} \approx 11.5 \text{ m}$ (b) At the sea, the displacement relative to the launch point is $s = -40$ m. Use $s = ut + \frac{1}{2}at^2$: $-40 = 15t + \frac{1}{2}(-9.8)t^2 \Rightarrow -40 = 15t - 4.9t^2 \Rightarrow 4.9t^2 - 15t - 40 = 0$ Solve via quadratic formula.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify the trigonometric expression $\frac{\sec x - \cos x}{\tan x}$.

a. $\sin x$ ✓

b. $\cos x$

c. $\cot x$

d. $\csc x$

$\sec x - \cos x = \frac{1}{\cos x} - \cos x = \frac{1 - \cos^2 x}{\cos x} = \frac{\sin^2 x}{\cos x}$. Dividing by $\tan x = \frac{\sin x}{\cos x}$ gives $\frac{\sin^2 x}{\cos x} \cdot \frac{\cos x}{\sin x} = \sin x$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the exact value of $\csc\left(\frac{5\pi}{6}\right)$.

a. 2 ✓

b. -2

c. $\frac{2}{\sqrt{3}}$

d. $-\frac{2}{\sqrt{3}}$

$\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$. Since $\csc\theta = \frac{1}{\sin\theta}$, $\csc\left(\frac{5\pi}{6}\right) = \frac{1}{1/2} = 2$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which identity relates $\cot\theta$ and $\csc\theta$ for all defined angles?

a. $1+\cot^2\theta=\csc^2\theta$ ✓

b. $1+\csc^2\theta=\cot^2\theta$

c. $\cot^2\theta-\csc^2\theta=1$

d. $\csc\theta=1+\cot\theta$

Dividing $\sin^2\theta+\cos^2\theta=1$ by $\sin^2\theta$ yields the fundamental Pythagorean identity $1+\cot^2\theta=\csc^2\theta$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Use compound angle identities to find the exact value of $\cos(75^\circ)=\cos(45^\circ+30^\circ)$.

a. $\frac{\sqrt{6}-\sqrt{2}}{4}$ ✓

b. $\frac{\sqrt{6}+\sqrt{2}}{4}$

c. $\frac{\sqrt{3}-1}{2\sqrt{2}}$

d. $\frac{\sqrt{2}-\sqrt{6}}{4}$

$\cos(A+B)=\cos A\cos B-\sin A\sin B=\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)-\left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)=\frac{\sqrt{6}-\sqrt{2}}{4}$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $\sin\theta=\frac{3}{5}$ and θ is acute, what is $\cos(2\theta)$?

a. $\frac{7}{25}$ ✓

b. $\frac{24}{25}$

c. $-\frac{7}{25}$

d. $\frac{16}{25}$

$\cos(2\theta)=1-2\sin^2\theta=1-2\left(\frac{3}{5}\right)^2=1-2\left(\frac{9}{25}\right)=1-\frac{18}{25}=\frac{7}{25}$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Under the same conditions ($\sin\theta=\frac{3}{5}$ acute), what is $\sin(2\theta)$?

a. $\frac{24}{25}$ ✓

b. $\frac{7}{25}$

c. $\frac{12}{25}$

d. $\frac{6}{5}$

$\cos\theta=\sqrt{1-9/25}=\frac{4}{5}$. Then $\sin(2\theta)=2\sin\theta\cos\theta=2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right)=\frac{24}{25}$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Express $\cos^2 x$ in terms of a first-power cosine function.

a. $\frac{1+\cos(2x)}{2}$ ✓

b. $\frac{1-\cos(2x)}{2}$

c. $\frac{1+\sin(2x)}{2}$

d. $1-\sin^2 x$

Rearranging $\cos(2x)=2\cos^2 x-1$ gives $2\cos^2 x=1+\cos(2x)\Rightarrow\cos^2 x=\frac{1+\cos(2x)}{2}$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Express $3\sin x + 4\cos x$ in the form $R\sin(x+\alpha)$ with $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

a. $5\sin(x + 0.927)$ ✓

b. $5\sin(x + 0.643)$

c. $7\sin(x + 0.927)$

d. $25\sin(x + 0.927)$

$R = \sqrt{3^2 + 4^2} = 5$. Expand $R\sin(x+\alpha) = R\cos\alpha\sin x + R\sin\alpha\cos x$. Thus $R\cos\alpha = 3$ and $R\sin\alpha = 4 \Rightarrow \tan\alpha = 4/3 \Rightarrow \alpha = \arctan(4/3) \approx 0.927$ rad.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the maximum value of the function $f(x) = 5\cos x - 12\sin x + 3$?

a. 16 ✓

b. 13

c. 10

d. 20

The combination $5\cos x - 12\sin x$ has amplitude $R = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = 13$. The maximum value is $13 + 3 = 16$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?

a. $\cos\theta$ ✓

b. $\sin\theta$

c. $\tan\theta$

d. 1

By definition, the coordinates of a point $P(x, y)$ on the unit circle corresponding to an angle θ (measured counterclockwise from the positive x-axis) are given by $(\cos\theta, \sin\theta)$. Therefore, the x-coordinate of the point P is $\cos\theta$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

a. $\frac{7}{25}$

b. $-\frac{7}{25}$ ✓

c. $\frac{24}{25}$

d. $-\frac{24}{25}$

We are given $\cos\alpha = \frac{3}{5}$ and α is in the first quadrant. Using the identity $\sin^2\alpha + \cos^2\alpha = 1$, we find $\sin\alpha = \sqrt{1 - (\frac{3}{5})^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$ (since α is in the first quadrant, $\sin\alpha$ is positive). We are given $\sin\beta = -\frac{4}{5}$ and β is in the third quadrant. Using the identity $\sin^2\beta + \cos^2\beta = 1$, we find.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$ ✓

d. $\frac{4}{5}$

On the unit circle, any point P(x, y) satisfies the equation $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $\frac{3}{5}$ and P is in the second quadrant. For a point $P(\cos\theta, \sin\theta)$ on the unit circle, we have $x = \cos\theta$ and $y = \sin\theta$. Thus, we have $\sin\theta = \frac{3}{5}$. Substituting this into the unit circle equation: $\cos^2\theta + (\frac{3}{5})^2 = 1$. $\cos^2\theta + \frac{9}{25} = 1$. $\cos^2\theta = 1 - \frac{9}{25}$. $\cos^2\theta = \frac{16}{25}$..

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

For any point on the unit circle, the coordinates are $(\cos(\theta), \sin(\theta))$. The equation of the unit circle is $x^2 + y^2 = 1$. Substituting the coordinates, we get $\cos^2(\theta) + \sin^2(\theta) = 1$. We are given that $\cos(\theta) = \frac{1}{2}$. So, $\left(\frac{1}{2}\right)^2 + \sin^2(\theta) = 1$. This simplifies to $\frac{1}{4} + \sin^2(\theta) = 1$. Subtracting $\frac{1}{4}$ from both sides gives $\sin^2(\theta) = 1 - \frac{1}{4} = \frac{3}{4}$. Taking the square root of both sides, we get.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$ ✓

d. 1

The coordinates of a point on the unit circle corresponding to an angle θ are given by $(\cos\theta, \sin\theta)$. In this case, $\theta = \frac{\pi}{3}$. Therefore, the y-coordinate of point P is $\sin\left(\frac{\pi}{3}\right)$. We know that $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{4}$

b. $\frac{1}{2}$ ✓

c. $\frac{3}{5}$

d. $-\frac{1}{2}$

$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}} = \sqrt{\frac{1-\left(\frac{1}{2}\right)}{2}} = \frac{1}{2}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the reference angle for 220°.

a. 60°

b. **40°** ✓

c. 50°

d. 140°

220° lies in Quadrant 3; the reference angle is 40°.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The table below shows the results of a survey on preferred music genres among a group of students. What is the relative frequency of students who prefer rock music?

a. 0.20

b. **0.25** ✓

c. 0.30

d. 0.40

To find the relative frequency of students who prefer rock music, we first need to find the total number of students surveyed. The total number of students is the sum of students who prefer each genre: Pop (10) + Rock (15) + Hip-Hop (20) + Electronic (15) = 60 students. The number of students who prefer rock music is 15. The relative frequency is calculated as: (Number of students who prefer rock music) / (Total number of students). Relative Frequency (Rock) = 15 / 60 = 1/4 = 0.25.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

A small study was conducted on the number of hours students spent studying per week. The results are summarized in the frequency table below: | Hours Studied (h) | Frequency (f) | | :---------- | :---------- | | $0 \le h < 5$ | 12 | | $5 \le h < 10$ | 25 | | $10 \le h < 15$ | 18 | | $15 \le h < 20$ | 15 | What is the total number of students surveyed?

a. **60** ✓

b. 70

c. 65

d. 75

To find the total number of students surveyed, we need to sum the frequencies for each category of hours studied. Total students = Frequency($0 \le h < 5$) + Frequency($5 \le h < 10$) + Frequency($10 \le h < 15$) + Frequency($15 \le h < 20$) Total students = 12 + 25 + 18 + 15 Total students = 70 Therefore, the total number of students surveyed is 70.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.02 The Poisson Distribution

Evaluate $\int_0^2 xe^{2x}\,dx$ (in terms of e).

a. $\frac{3}{4}e^4 - \frac{1}{4}$

b. $\frac{3}{4}e^4$

c. $\frac{3}{4}e^4 + \frac{1}{4}$ ✓

d. $\frac{1}{4}e^4 + \frac{3}{4}$

Working through this calculus step: $\left[\frac{x}{2}e^{2x} - \frac{1}{4}e^{2x}\right]_0^2 = \frac{3}{4}e^4 + \frac{1}{4} - \left(0 - \frac{1}{4}\right) = \frac{3}{4}e^4 + \frac{1}{4}$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Further Integration Techniques

A 10-ft ladder leans against a wall; the base is 6 ft from the wall and slides away at $\frac{dx}{dt} = 1$ ft/s. Find $\frac{dy}{dt}$, the rate the top slides down (height $y = 8$ ft).

a. -1 ft/s

b. $-\frac{3}{4}$ ft/s ✓

c. $-\frac{1}{4}$ ft/s

d. $-\frac{1}{2}$ ft/s

Working through this calculus step: $x^2 + y^2 = 10^2 \Rightarrow 2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt} = -\frac{6}{8}(1) = -\frac{3}{4}$

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.02 Differential Equations

A graph has edges $1-2, 2-3, 3-4, 4-1$. Does it have an Euler circuit, an Euler trail (not circuit), or neither?

a. Euler circuit exists ✓

b. Euler trail exists, but not an Euler circuit

c. Cannot be determined from the given information

d. Neither an Euler trail nor circuit exists

A connected graph has an Euler circuit iff every vertex has even degree; an Euler trail (not circuit) iff exactly 2 vertices have odd degree.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

A weighted graph has edges (with weights) $1{-}2(1), 2{-}3(6), 3{-}4(5), 4{-}5(7), 1{-}5(8), 2{-}5(2), 1{-}3(9)$. Find the total weight of the minimum spanning tree.

a. 18

b. 11

c. 10

d. **14** ✓

Applying Kruskal's algorithm (adding edges in increasing weight order, skipping cycles) gives a minimum spanning tree of total weight 14.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{2}{5}, \frac{2}{5}, \frac{3}{10}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

a. 0.383

b. 0.533

c. **0.333** ✓

d. 0.433

$P(\text{def}) = 0.360$ (law of total probability), $P(M_2|\text{def}) = \frac{P(M_2)P(\text{def}|M_2)}{P(\text{def})} \approx 0.333$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Matrices (AI HL)

Given $P(A) = \frac{2}{5}$, $P(B|A) = \frac{1}{2}$, and $P(B|A') = \frac{3}{10}$, find $P(A|B)$ to 3 decimal places.

a. 0.426

b. 0.576

c. **0.526** ✓

d. 0.476

Working through this statistics step: $P(B) = P(A)P(B|A) + P(A')P(B|A') = 0.40 \times 0.50 + 0.60 \times 0.30 = 0.380.$

$P(A|B) = \frac{P(A)P(B|A)}{P(B)} \approx 0.526$

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

If $\frac{dy}{dt} = \frac{1}{4}y$ and $y(0) = 9$, find $y(4)$ in terms of e.

a. $9 + e$

b. $9e^{-1}$

c. $9e$ ✓

d. $9\frac{1}{4}e$

Working through this calculus step: $\frac{dy}{y} = \frac{1}{4}dt \Rightarrow y = Ae^{\frac{1}{4}t},\ y(0) = 9 \Rightarrow A = 9 \Rightarrow y(4) = 9e$

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